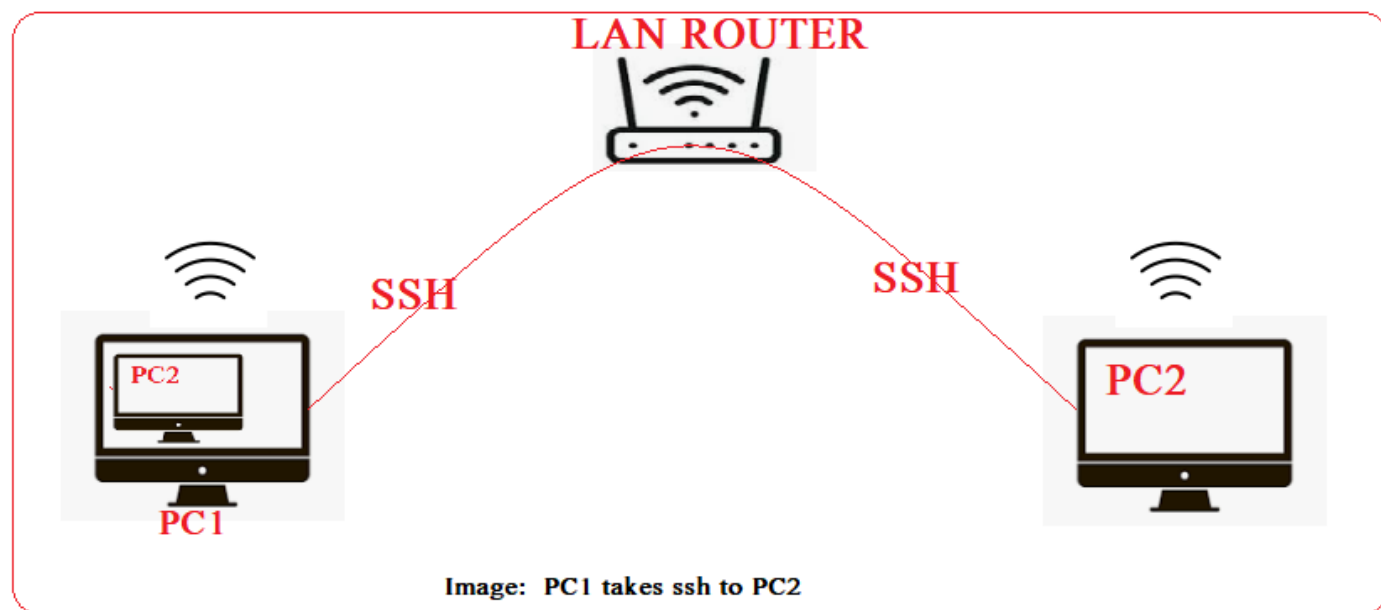


➤ WHAT IS SSH? (SSH OVERVIEW?)

SSH, or Secure Shell, is a protocol used to securely connect to another computer over a network. It allows you to remotely log in, manage files, run commands, and even transfer data, all while keeping your information encrypted and safe from hacker's eyes. SSH can be used by anyone including system administrators and developers to manage servers and network devices securely over the network.



➤ WHAT IS REMOTE LOGIN?

Remote login is a way of accessing a computer from a different location over the network as if you were sitting right in front of it. With SSH, you can log into a remote server or another system, run commands, and manage it from anywhere, as long as you have network access. It's especially helpful when working with servers or systems that don't have physical monitors or keyboards attached.

➤ WHAT IS ADVANTAGES OF SSH?

SSH provides several key benefits:

- **Security:** It encrypts all your data and communication, so your login details and commands stay private, preventing a "middle-man" from being able to see or intercept your communication.
- **Secure Authentication:** You can log in using either a password or a more secure key-based method.
- **Data Integrity:** SSH ensures the data sent between your computer and the remote system isn't compromised by anyone, maintaining its integrity during transmission.
- **Remote Command Execution:** You can run commands on a remote server just as if you were using it locally, making it easier to manage distant systems.
- **Secure File Transfer:** SSH lets you transfer files between computers safely using tools like SCP or SFTP, ensuring the data is encrypted during the process.

- **Port Forwarding:** You can securely tunnel other services or connections through your SSH session, making it more versatile. This means you can create your secret channel to access systems or services securely.

➤ HOW TO SET UP SSH (steps for installing and configuring SSH on a system.)

Step-1 (install openssh server package)

First, ensure that OpenSSH is installed on your system. You can install it using your package manager tool like yum & dnf:

```
[root@rhcsa-server ~]# yum install -y openssh
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use subscription-manager to register.

Last metadata expiration check: 0:21:00 ago on Thursday 19 September 2024 13:17:14 IST.
Package openssh-8.7p1-24.el9_1.x86_64 is already installed.
Dependencies resolved.
=====
Package                                Architecture                Version
=====
Upgrading:
openssh                                x86_64                       8.7p1-43.el9
openssh-clients                        x86_64                       8.7p1-43.el9
openssh-server                        x86_64                       8.7p1-43.el9
Transaction Summary

```

Step-2: Start and enable SSH service

Once installed, start the SSH service and ensure it starts automatically on boot.

```
[root@rhcsa-server ~]# systemctl enable --now sshd
Created symlink /etc/systemd/system/multi-user.target.wants/sshd.service → /usr/lib/systemd/system/sshd.service.
[root@rhcsa-server ~]#

[root@rhcsa-server ~]# systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2024-09-19 13:17:14 IST; 58min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Main PID: 6826 (sshd)
    Tasks: 1 (limit: 22832)
   Memory: 1.7M
      CPU: 27ms
   CGroup: /system.slice/sshd.service
           └─6826 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 19 13:17:14 rhcsa-server.example.com systemd[1]: Starting OpenSSH server daemon...
Sep 19 13:17:14 rhcsa-server.example.com sshd[6826]: Server listening on 0.0.0.0 port 22.
Sep 19 13:17:14 rhcsa-server.example.com sshd[6826]: Server listening on :: port 22.
Sep 19 13:17:14 rhcsa-server.example.com systemd[1]: Started OpenSSH server daemon.
[root@rhcsa-server ~]#
```

Step-3: Allow SSH through the firewall

If your system has a firewall, you need to allow SSH traffic. Here's how you can open port 22 (the default SSH port):

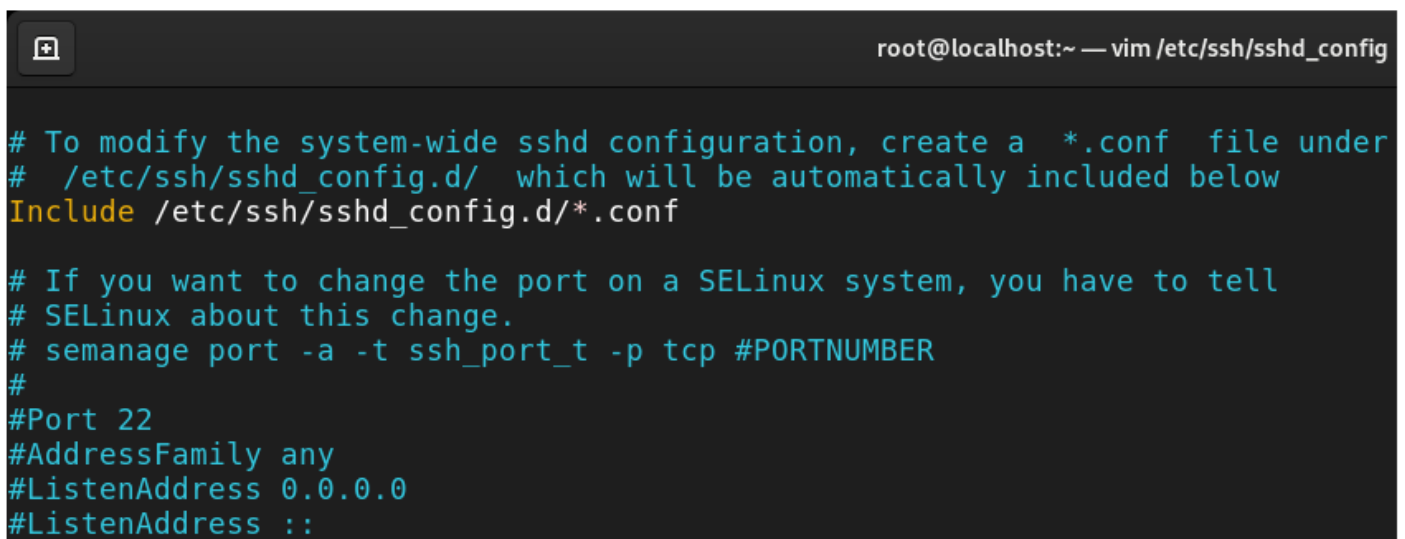
```
[root@rhcsa-server ~]# firewall-cmd --permanent --add-port=22/tcp
success
[root@rhcsa-server ~]# firewall-cmd --reload
success
[root@rhcsa-server ~]#
```

Step-4: Configure SSH (*optional)

You can modify the SSH configuration file to improve security or change its default behavior. The configuration file is located at /etc/ssh/sshd_config. For this open the configuration file and some key options you might want to modify:

Step-4.1: (open the configuration file)

```
[root@rhcsa-server ~]# vim /etc/ssh/sshd_config
```



```
# To modify the system-wide sshd configuration, create a *.conf file under
# /etc/ssh/sshd_config.d/ which will be automatically included below
Include /etc/ssh/sshd_config.d/*.conf

# If you want to change the port on a SELinux system, you have to tell
# SELinux about this change.
# semanage port -a -t ssh_port_t -p tcp #PORTNUMBER
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::
```

Step-4.2: (Change the default SSH port for added security.)

```
# semanage port -a -t ssh_port_t -p tcp #PORTNUMBER

Port 2224 #Change the default SSH port for added security
```

Step-4.3: (PermitRootLogin: Disable root login to enhance security.)

```
PermitRootLogin prohibit-password
#PermitRootLogin yes
```

Step-4.4 (Allow or Deny SSH access to specific users & Groups.)

```
#Example to allow specific users:
AllowUsers user1 user2

#Example to Deny specific users:
DenyUsers user3 user4

#Example to allow specific groups users:
AllowGroups sshusers admin

#Example to Deny specific groups users:
DenyGroups restricted nogroup
```

Step-5: Modifies the SELinux (security-enhanced Linux) policy to allow the ssh service to use a non-default port (in this case, port 2224).

```
[root@rhcsa-server ~]# semanage port -a -t ssh_port_t -p tcp 2224
```

Step-6: Restart the ssh service (*mandatory to apply the changed)

```
[root@rhcsa-server ~]# systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2024-09-19 15:15:33 IST; 3min 21s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Main PID: 8258 (sshd)
    Tasks: 1 (limit: 22832)
   Memory: 1.7M
      CPU: 19ms
   CGroup: /system.slice/sshd.service
           └─8258 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

Sep 19 15:15:33 rhcsa-server.example.com systemd[1]: Starting OpenSSH server daemon...
Sep 19 15:15:33 rhcsa-server.example.com sshd[8258]: Server listening on 0.0.0.0 port 2224.
Sep 19 15:15:33 rhcsa-server.example.com sshd[8258]: Server listening on :: port 2224.
Sep 19 15:15:33 rhcsa-server.example.com systemd[1]: Started OpenSSH server daemon.
[root@rhcsa-server ~]#

An orange speech bubble with the word "Info" inside, pointing towards the systemctl status output.

➤ Step-6 Verify SSH connection (test SSH connectivity from another system)

```
[root@server2 ~]# ssh admin@192.168.199.172 -p 2224
The authenticity of host '[192.168.199.172]:2224 ([192.168.199.172]:2224)' can't be established.
ED25519 key fingerprint is SHA256:h2FjDWYc1v2/jGyTkksAt+J2lLKvyOforiU6BxyevbY.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[192.168.199.172]:2224' (ED25519) to the list of known hosts.
admin@192.168.199.172's password:
Activate the web console with: systemctl enable --now cockpit.socket

Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
[admin@server2 ~]$ whoami
admin
```

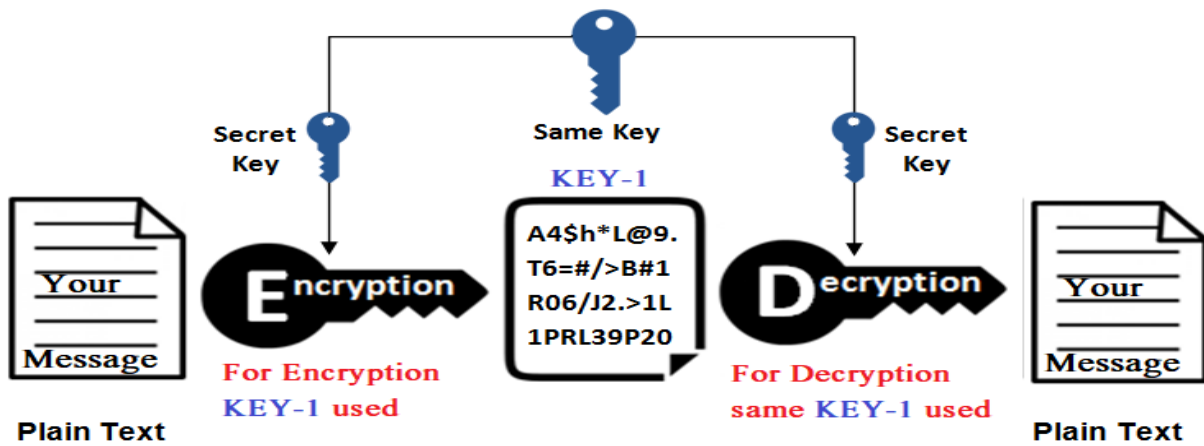
➤ What is encryption? Types of encryption [symmetric vs. Asymmetric]

Encryption is the process of converting plain text data into an unreadable format to protect it from unauthorized access. This ensures that sensitive information, like passwords or private messages, stays secure when being sent over a network or stored. Only those with the right key can turn the encrypted data back into its original, readable form.

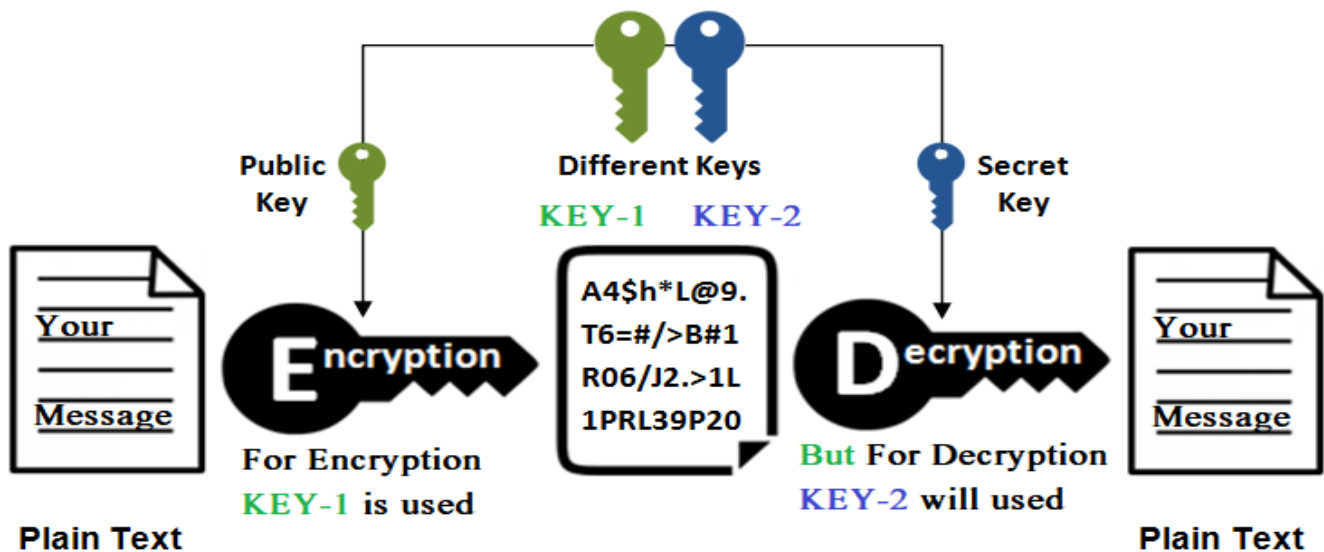
Symmetric vs. Asymmetric key

- **Symmetric Key Encryption:** Uses a single key for both encryption and decryption. Both the sender and receiver use the same key to encode and decode the data. It's fast and efficient but requires that both parties securely share the key in advance. This is less secure.
- **Asymmetric Key Encryption:** Uses a pair of keys – a public key and a private key. The public key is used for encryption, and the private key is used for decryption. This allows secure communication without needing to share a key in advance. The public key can be shared openly, while the private key is kept secret. This is a secure method.

Symmetric Encryption



Asymmetric Encryption



- How to log in remotely with ssh, ssh command's options
- **Basic Login:** (ssh username@hostname) or (ssh username@ip_address)

```
[root@server2 ~]# ssh admin@192.168.199.172
admin@192.168.199.172's password:
Activate the web console with: systemctl enable --now cockpit.socket

Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Thu Sep 19 16:32:38 2024 from 192.168.199.172
[admin@server2 ~]$
```

```
[root@server2 ~]# ssh admin@localhost
admin@localhost's password:
Activate the web console with: systemctl enable --now cockpit.socket

Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Thu Sep 19 16:36:27 2024 from ::1
[admin@server2 ~]$
```

- **Login with a Specific Port:** (ssh username@hostname -p 2224)

```
[root@server2 ~]# ssh admin@192.168.199.172 -p 2224
admin@192.168.199.172's password:
Activate the web console with: systemctl enable --now cockpit.socket

Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Thu Sep 19 15:27:06 2024 from 192.168.199.172
[admin@server2 ~]$
```

➤ **Login with a Specific Private Key: (ssh username@hostname -i ~/.ssh/id_rsa)**

```
[root@server2 ~]# ssh admin@192.168.199.172 -i ~/.ssh/id_rsa
Activate the web console with: systemctl enable --now cockpit.socket

Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Thu Sep 19 16:51:02 2024 from 192.168.199.172
[admin@server2 ~]$
```

➤ **Login with more detailed debugging Output: (ssh -vvv username@hostname)**

```
[root@server2 ~]# ssh -vvv admin@192.168.199.172
OpenSSH_8.7p1, OpenSSL 3.0.1 14 Dec 2021
debug1: Reading configuration data /etc/ssh/ssh_config
debug3: /etc/ssh/ssh_config line 55: Including file /etc/ssh/ssh_config.d/50-redhat.conf
debug1: Reading configuration data /etc/ssh/ssh_config.d/50-redhat.conf
debug2: checking match for 'final all' host 192.168.199.172 originally 192.168.199.172
debug3: /etc/ssh/ssh_config.d/50-redhat.conf line 3: not matched 'final'
debug2: match not found
debug3: /etc/ssh/ssh_config.d/50-redhat.conf line 5: Including file /etc/crypto-policies
(parse only)
debug1: Reading configuration data /etc/crypto-policies/back-ends/openssh.config
debug3: gss kex names ok: [gss-curve25519-sha256-,gss-nistp256-sha256-,gss-group14-sha256-]
debug3: kex names ok: [curve25519-sha256,curve25519-sha256@libssh.org,ecdh-sha2-nistp256,
p521,diffie-hellman-group-exchange-sha256,diffie-hellman-group14-sha256,diffie-hellman-
18-sha512]
debug1: configuration requests final Match pass
debug3: resolve_canonicalize: hostname 192.168.199.172 is address
```

➤ **Exiting the SSH Session: (To exit the SSH session, simply type exit or press Ctrl+D.)**

```
[root@server2 ~]# ssh admin@192.168.199.172
Activate the web console with: systemctl enable --now cockpit.socket

Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Thu Sep 19 16:48:25 2024
[admin@server2 ~]$ exit
logout
Connection to 192.168.199.172 closed.
[root@server2 ~]#
```

➤ **Password-less authentication overview**

Password-less authentication is a method of logging into a remote system without needing to manually enter a password each time. It is commonly implemented using SSH key-based authentication, which uses a cryptographic key pair (public and private keys) instead of passwords for authentication.

How It Works:

1. **Generate Key Pair: (ssh-keygen)**

- A key pair consists of:

- Private Key: Stored securely on the local machine.
- Public Key: Shared with the remote server.
- Use ssh-keygen to create this pair.

```
[root@server2 ~]# ssh-keygen
Generating public/private rsa key pair.

Enter file in which to save the key (/root/.ssh/id_rsa): Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:t3iGAbCenTcq4dYt4XdCbX1cNcdnL9G2BXCaaTG7N20 root@server2
The key's randomart image is:
+---[RSA 3072]---+
| .          +.o=o |
| o          0..X |
| . .        * .+= |
| . o o    o o.o. |
| + + S . + o.E |
| . + * B o o o |
| + = * B . |
| . . o * |
+-----[SHA256]-----+
[root@server2 ~]#
```

2. Copy Public Key to Remote Server: (ssh-copy-id username@hostname)

- The public key is copied to the remote server's ~/.ssh/authorized_keys file using ssh-copy-id or manually via scp or other file transfer methods.
- The private key remains on your local machine.

```
[root@server2 ~]# ssh-copy-id admin@192.168.199.172
The authenticity of host '192.168.199.172 (192.168.199.172)' can't be established.
ED25519 key fingerprint is SHA256:h2FjDWYclv2/jGyTkksAt+J2lLKvy0foriU6BxyevbY.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [192.168.199.172]:2224
  ~/.ssh/known_hosts:4: [localhost]:2224
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now
admin@192.168.199.172's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'admin@192.168.199.172'"
and check to make sure that only the key(s) you wanted were added.
```

3. Authentication Process:

- When you attempt to connect to the remote server, the SSH client on your machine uses your private key to prove your identity.
- The server verifies your identity by checking if your public key matches the private key.
- If the keys match, access is granted without prompting for a password.

```
[root@server2 ~]# ssh admin@192.168.199.172
Activate the web console with: systemctl enable --now cockpit.socket

Register this system with Red Hat Insights: insights-client --register
Create an account or view all your systems at https://red.ht/insights-dashboard
Last login: Thu Sep 19 16:48:25 2024
[admin@server2 ~]$
```